The Lucid Beamer Theme

Every slide shows the code that made it

A copy-and-paste manual

Outline

Start a deck: two lines

- Put `beamerthemelucid.sty` next to your `.tex` file

```
\documentclass[11pt,aspectratio=43]{beamer}
\usetheme{lucid}
```

- Use `aspectratio=169` for widescreen

Compile twice. The outline slides need a second pass. snippets/start.tex

The title slide I

- **No name, institution or logo appears unless you set one**
- `[plain]` drops the footer, which a title slide should not have

```
\title{Your Title}
\subtitle{An optional second line}
\lucidevent{Venue or occasion}
\date{}
% \author{Your Name}             % optional
% \institute{Your Institution}  % optional
% \lucidlogo{logo.pdf}          % optional

\begin{frame}[plain]
  \titlepage
\end{frame}
```

snippets/title-slide.tex

Make a new slide I

- A slide is a **frame**. Title in braces, content between
- Leave the braces off for a slide with no title

```
% A slide is a "frame". This is the whole pattern:

\begin{frame}{The slide title}
  \begin{itemize}
    \item A point
    \item Another point
  \end{itemize}
\end{frame}

% No title? Leave the braces off:
\begin{frame}
  \takeaway{A single sentence, no title.}
\end{frame}

% No title bar, no footer -- for a full-bleed image or a cover:
\begin{frame}[plain]
  \slidefigure[\paperwidth]{cover.pdf}
```

Make a new slide II

```
\end{frame}

% Reveal the bullets one at a time:
\begin{frame}{Built up in steps}
  \begin{itemize}
    \item<1-> Appears first
    \item<2-> Appears second
  \end{itemize}
\end{frame}
```

snippets/new-slide.tex

Hide a slide I

- Comment it out to drop it for now
- Move it below `backupframes` to keep it without counting it

```
% Wrap anything you want to park. Whole frames are fine.

\begin{hidden}
\begin{frame}{Cut for time}
  \begin{itemize}
    \item Still here, just not in the deck
  \end{itemize}
\end{frame}
\end{hidden}

% Then choose a mode in the preamble. Two, and they are opposites:

\usetheme{lucid}                  % hidden=off is the default:
                                  % the block is NOT compiled. Not in
                                  % the PDF, not in the page count.
```

Hide a slide II

```
\usetheme[hidden=show]{lucid}     % compile them, so you can see what
                                  % you have parked.

% Better than commenting out, which is tedious to undo and invisible to
% a search. Better than deleting, which loses the slide.

% --- Two smaller variants -----------------------------------------

% Keep it in the deck but out of the count -- put it below \backupframes:
\backupframes

\begin{frame}{Held in reserve}
  ...
\end{frame}

% Keep it visible but unnumbered:
\begin{frame}[noframenumbering]{Not counted}
  ...
\end{frame}
```

snippets/hide-slide.tex

Speaker notes I

- `note` sits inside the frame it belongs to
- Nothing is shown unless you ask for it, so notes are safe to leave in

```
% Write the note inside the frame it belongs to:

\begin{frame}{Results}
  \begin{itemize}
    \item The gain holds across both attackers
  \end{itemize}
  \speakernote{Say the number out loud: 78 against 62.
    If asked about variance, point at the backup slide.}
\end{frame}

% Then choose a mode in the preamble. Two matter, and they are opposites:

\usetheme{lucid}                 % notes=off is the default:
                                 % notes are NOT compiled at all.
                                 % This is the file you hand out.
```

Speaker notes II

```
\usetheme[notes=pages]{lucid}     % each slide is followed by a typeset
                                  % notes page, in the deck's own colours.

% Two more, for presenting:
\usetheme[notes=only]{lucid}      % notes without slides -- a script
\usetheme[notes=second]{lucid}    % presenter view, notes on screen two

% Beamer's own \note{...} works too and obeys the same option.
% Tip: build twice -- once plain to present from, once with
% notes=pages to print.
```

snippets/notes.tex

Animation: one slide, several pages

- Each step prints as **another page** of the PDF

This slide is three pages. In handout mode it collapses back to one.

Animation: one slide, several pages

- Each step prints as **another page** of the PDF
- Nothing moves — the reader sees a page at a time

This slide is three pages. In handout mode it collapses back to one.

Animation: one slide, several pages

- Each step prints as **another page** of the PDF
- Nothing moves — the reader sees a page at a time
- So an animated deck has more pages than slides

This slide is three pages. In handout mode it collapses back to one.

Animation: the syntax I

```
% An "animation" here is not motion. Each step prints as ANOTHER PAGE
% of the PDF, showing a little more. A frame with three steps becomes
% three pages. That is why an animated deck has more pages than slides.

% --- Bullets, one at a time ---------------------------------------
\begin{frame}{Built up}
  \begin{steps}
    \item Appears on page 1
    \item Appears on page 2
    \item Appears on page 3
  \end{steps}
\end{frame}

% --- Break anywhere ------------------------------------------------
\begin{frame}{Two halves}
  The setup.
  \stepafter
  The punchline.
\end{frame}

% --- Precise control -----------------------------------------------
\showfrom{2}{Appears from page 2 on; space held before it.}
\showonly{2}{Appears only on page 2; no space held.}
```

Animation: the syntax II

```
\emphat{2}{Highlighted on page 2, plain otherwise.}
\swap{2}{Before}{After}

% --- Swapping a figure in place ------------------------------------
\begin{frame}{Same spot, two figures}
  \showonly{1}{\slidefigure{before.pdf}}%
  \showonly{2}{\slidefigure{after.pdf}}%
\end{frame}

% Beamer's own <1->, \pause, \only and \alt still work. Use whichever
% reads better. Keep animation for places where the ORDER matters --
% every step is a page someone has to sit through.

% --- A figure that stays while the text builds ---------------------
% The classic trap: put the overlays INSIDE a pane, not around the
% columns, or the whole split appears only on the last page.
\begin{frame}{Figure holds, text builds}
  \begin{figuretext}[0.55]
    \begin{steps}
      \item First point
      \item Second point
    \end{steps}
    \nextpane
```

Animation: the syntax III

```
    \slidefigure{diagram.pdf}   % no overlay spec: present throughout
  \end{figuretext}
\end{frame}

% Printing an animated deck? See snippets/handout.tex -- one word
% collapses every step back to a single page per slide.
```

snippets/animation.tex

Printing an animated deck I

- **handout** collapses every step back to one page per slide
- Build three files from one body: the talk, the handout, your script

Printing an animated deck II

```
% A deck with animation is painful to print: every step is its own page,
% so a 20-slide talk becomes 60 pages. "handout" flattens that -- each
% frame collapses back to ONE page with everything shown.

\documentclass[11pt,aspectratio=43,handout]{beamer}
\usetheme{lucid}

% That single word turns off every overlay: \steps, \stepafter,
% \showfrom, \showonly, \swap and beamer's own <1-> all stop splitting.

% --- Watch out -----------------------------------------------------
% \showonly{n}{...} reserves no space, so two \showonly blocks that
% shared a spot in the talk land ON TOP of each other in a handout.
% If a slide swaps one figure for another, use \swap instead:
\swap{2}{\slidefigure{before.pdf}}{\slidefigure{after.pdf}}

% --- The three files worth building -------------------------------
% 1. the talk        \documentclass[...]{beamer}
% 2. the handout     \documentclass[...,handout]{beamer}
% 3. your script     \usetheme[notes=only]{lucid}

% --- Four slides to a page, for printing --------------------------
```

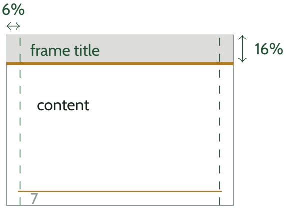
Printing an animated deck III

```
\usepackage{pgfpages}
\pgfpagesuselayout{4 on 1}[letterpaper,landscape,border shrink=5mm]

% Keep the talk and the handout as separate .tex files, each a couple of
% lines long, both \input-ing the same body. Then they cannot drift.
```

snippets/handout.tex

Outline

The grid, and what you can move I



- Rail, band and footer are **fixed by the theme**
- You place content inside them; you do not set margins
- Content starts under the rule and runs down

```
% The grid is fixed by the theme. You do not set margins; you place
% content inside them.
%
%   left/right rail    6.0% of page width   title and body share it
%   title band        16.0% of page height  full bleed, grey
%   rule under band    0.67% of page height full bleed, amber
%   footer rule       92.6% down the page   slide number, date
```

The grid, and what you can move II

```
%
% Content starts under the rule and runs down. Useful controls:

% Push content to the bottom of the content area:
\vfill
\takeaway{The last thing they should see.}

% Split the slide. The number is the LEFT pane's share:
\begin{figuretext}[0.55]
  \begin{itemize}
    \item Left pane
  \end{itemize}
  \nextpane
  \slidefigure{diagram.pdf}
\end{figuretext}

% Centre this one frame instead of top-aligning it:
\begin{frame}[c]{A centred slide}
  \takeaway{One line, centred.}
\end{frame}

% Drop the chrome entirely for a full-bleed slide:
\begin{frame}[plain]
```

The grid, and what you can move III

```
  \slidefigure[\paperwidth]{cover.pdf}
\end{frame}

% Deck-wide switches:
%   \usetheme[align=center]{lucid}   centre every slide
%   \usetheme[footer=none]{lucid}    keep the rule, drop the text
```

snippets/layout.tex

Outline

Getting started

The layout

Text

Figures

Tables, maths, closings

Options and backup

Bullets: two levels, and a third to avoid

- Level one
  - Level two
    - Level three is muted on purpose

```
\begin{itemize}
  \item Level one
    \begin{itemize}
      \item Level two
    \end{itemize}
\end{itemize}
```

Aim for two top-level bullets and 25–30 words a slide.   snippets/bullets.tex

Emphasis: three devices, and none of them is bold

- A **primary term**, a *secondary term*, something highlighted
- Each one survives greyscale printing on its own

```
A \term{primary term}, a \altterm{secondary term},
something \hl{highlighted}
```

snippets/emphasis.tex

Maths beyond one equation I

- Aligned lines, cases, matrices, and numbering when you need it

$$\begin{aligned}\mathcal{L} &= \mathbb{E}_\pi\left[\sum_t \gamma^t r_t\right] \\ &= \sum_s d^\pi(s)\, v^\pi(s)\end{aligned}$$

$$r(s) = \begin{cases} +1 & \text{host recovered} \\ -1 & \text{host compromised} \\ 0 & \text{otherwise} \end{cases}$$

Maths beyond one equation II

```
% amsmath is already loaded by the theme. So is a sans maths font, so
% symbols match the body weight instead of sitting in a serif face.

% --- Inline and display ------------------------------------------
The error level $\lambda$ controls it.

\[ q_j = \frac{e^{\lambda \mathrm{EU}_j(x)}}{\sum_i e^{\lambda \mathrm{EU}_i(x)}} \]

% Display maths is UNNUMBERED by default. A talk has no cross-references.

% --- When you must number one ------------------------------------
\begin{equation}
  \mathcal{L} = \mathbb{E}_{\pi}\left[ \sum_t \gamma^t r_t \right]
  \label{eq:objective}
\end{equation}
Referred to later as \eqref{eq:objective}.

% --- Several lines, aligned on = ---------------------------------
\begin{align*}
  \mathcal{L} &= \mathbb{E}_{\pi}\left[ \sum_t \gamma^t r_t \right] \\
              &= \sum_s d^{\pi}(s) \, v^{\pi}(s)
\end{align*}
```

Maths beyond one equation III

```
% align* is unnumbered; align numbers every line.

% --- Cases and matrices ------------------------------------------
\[ r(s) = \begin{cases}
     +1 & \text{host recovered} \\
     -1 & \text{host compromised} \\
      0 & \text{otherwise}
   \end{cases} \]

\[ A = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \]

% --- Pointing at part of an equation ------------------------------
% Colour the piece you are talking about:
\[ q_j = \frac{e^{\tint{accent}{\lambda} \mathrm{EU}_j(x)}}
              {\sum_i e^{\lambda \mathrm{EU}_i(x)}} \]

% Or reveal it, then emphasise it, across two pages:
\begin{frame}{Building the objective}
  \[ \mathcal{L} = \emphat{2}{\mathbb{E}_{\pi}}\left[ \sum_t \gamma^t r_t \right] \]
  \showfrom{2}{The expectation is over the policy, not the data.}
\end{frame}
```

Maths beyond one equation IV

```latex
% --- Statements ------------------------------------------------------
\begin{theorem}
  The equilibrium exists for every finite game.
\end{theorem}

\begin{definition}
  A \term{world model} predicts the next observation from the last.
\end{definition}

% Also available: lemma, corollary, proof, example, block.
% Give any of them a name: \begin{theorem}[Nash, 1951]

% --- Two habits worth keeping ----------------------------------------
% \mathrm{} for words inside maths: \mathrm{EU}, not EU
% \dfrac for a full-size fraction inside inline maths
```

snippets/math.tex

Statements

Definition

A **world model** predicts the next observation from the last.

Theorem (Nash, 1951)

Every finite game has an equilibrium in mixed strategies.

- Also: `lemma`, `corollary`, `proof`, `example`, `block`

They take the deck's colours, and their bodies stay upright to read from the back.

Pointing at one part of an equation

$$q_j = \frac{e^{\lambda \, \mathrm{EU}_j(x)}}{\sum_i e^{\lambda \, \mathrm{EU}_i(x)}}$$

- Colour the piece you are talking about, or reveal it with `emphat` and `showfrom`

Colour, highlight, underline I

- **a primary term**, *a secondary term*, highlighted, underlined, alerted
- Or reach into the palette directly: gold, muted, teal

```
% Reach for these first. They are the deck's own vocabulary, so a whole
% talk stays consistent and every one survives greyscale printing.

\term{a primary term}        % accent, bold
\altterm{a secondary term}   % teal, italic
\hl{highlighted}             % amber wash behind the text
\ul{underlined}              % plain underline
\alert{alerted}              % beamer's own, mapped to the accent

% --- Arbitrary colour, still from the palette ---------------------
\tint{accent}{text}   % forest green -- structure and emphasis
\tint{gold}{text}     % amber        -- rules and second-level marks
\tint{muted}{text}    % grey-green   -- deprioritised text
\tint{ink}{text}      % near-black   -- body
\tint{termA}{text}    % forest       -- first defined term
```

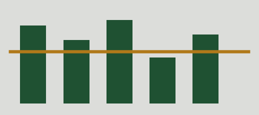
Colour, highlight, underline II

```
\tint{termB}{text}    % teal         -- second defined term

% --- Raw colour, if you really need it -----------------------------
{\color{lucidAccent} some words}
\textcolor{lucidGold}{some words}

% Palette names you can use anywhere, including in tikz and tables:
%   lucidInk lucidAccent lucidGold lucidMuted lucidBand lucidWash
%   lucidTermA lucidTermB

% --- Colouring part of a table or figure ---------------------------
Ours & \term{78} \\
\rowcolor{lucidBand} header & row \\
\fill[lucidAccent] (0,0) rectangle (1,2);   % inside tikz

% Restraint is the point: the reference decks use three or four colours
% for ~95% of all text. Bold is about 3%. If everything is emphasised,
% nothing is.
```

snippets/color.tex   Three or four colours carry 95% of the text in a good deck.

Outline

A figure, uncaptioned



```
\slidefigure{plot.pdf}
\slidefigure[0.9\textwidth]{plot.pdf}   % wider or narrower
\figcaption{Only if a caption is truly needed.}
```

snippets/figure.tex

A figure beside text

- The left pane holds your points
- The argument sets its share

```
\begin{figuretext}[0.55]
  \begin{itemize}
    \item Your points
  \end{itemize}
  \nextpane
  \slidefigure{diagram.pdf}
\end{figuretext}
```

snippets/figure-text.tex

Outline

A table

Method	Score
Baseline	62
Ours	**78**

```
\begin{lucidtable}
\begin{tabular}{lr}
  \toprule
  \thead{Method & Score} \\
  \midrule
  Baseline & 62 \\
  Ours     & \term{78} \\
  \bottomrule
\end{tabular}
\end{lucidtable}
```

Horizontal rules only. snippets/table.tex

An equation

- Equations are unnumbered, and sit inside the bullet that introduces them

$$q_j = \frac{e^{\lambda \mathrm{EU}_j(x)}}{\sum_i e^{\lambda \mathrm{EU}_i(x)}}$$

```
\begin{itemize}
  \item Probability of attacking target $j$
    \[ q_j = \frac{e^{\lambda \mathrm{EU}_j(x)}}
                  {\sum_i e^{\lambda \mathrm{EU}_i(x)}} \]
\end{itemize}
```

snippets/equation.tex

A source line, and a closing statement

The one sentence you want them to leave with.

```
\takeaway{The one sentence you want them to leave with.}
\citeline{Author, A. (2020). Title. Venue.}
```

A source line sits just above the footer rule, like this one.

Code on a slide I

- A frame holding verbatim text must be `[fragile]`

```python
for step in range(n_steps):
    loss = model.update(batch)
```

```latex
% In the preamble, once:
\usepackage{listings}
\lstset{
  basicstyle=\ttfamily\footnotesize\color{lucidInk},
  keywordstyle=\color{lucidAccent},
  commentstyle=\color{lucidMuted}\itshape,
  backgroundcolor=\color{lucidBand!40},
  breaklines=true, columns=fullflexible,
  xleftmargin=5pt, frame=none, showstringspaces=false,
}

% On a slide. The [fragile] is required for verbatim text:
\begin{frame}[fragile]{Training loop}
\begin{lstlisting}[language=Python]
for step in range(n_steps):
    loss = model.update(batch)
```

Code on a slide II

```
\end{lstlisting}
\end{frame}

% Long or awkward code is safer in its own file:
\lstinputlisting[language=Python]{code/train.py}

% Trap: a [fragile] frame ends at the first \end{frame} at the start of
% a line -- including one that is only sample text inside a listing.
% If you must show \end{frame}, use \lstinputlisting from a file.
```

snippets/code.tex

Citing sources I

- `citeline` is the light option: a credit line, no `.bib` needed
- For real citations, use a `.bib` and cite as usual [1]

```
% --- 1. a light source line, no .bib file needed ------------------
\citeline{McKelvey \& Palfrey (1995). Games and Economic Behavior.}

% --- 2. real citations from a .bib file ---------------------------
% Cite in the text:
Quantal response \cite{mckelvey1995quantal} explains it.

% Then one frame at the end of the deck:
\begin{frame}[allowframebreaks]{References}
  \footnotesize
  \bibliographystyle{plain}
  \bibliography{refs}
\end{frame}

% Build order: pdflatex, bibtex, pdflatex, pdflatex.
```

Citing sources II

```
% Overleaf and latexmk do this for you.
```

Build order: pdflatex, bibtex, pdflatex, pdflatex.

Sections make their own divider

- Every `section` inserts an outline slide with the current section marked
- You have passed four of them already

```
\section{Your section name}

% To place the outline slide by hand instead:
%   \usetheme[outline=manual]{lucid}
%   \outlineframe
```

snippets/sections.tex

Outline

Theme options

Option	Values, default first
`palette`	`green`, `maroon`, `slate`
`font`	`cabin`, `helvet`, `none`
`footer`	`full`, `page`, `none`
`pagenumber`	`plain`, `total`
`outline`	`auto`, `manual`
`align`	`top`, `center`
`blind`	`false`, `true`

```
\usetheme[palette=maroon,pagenumber=total]{lucid}
```

Anonymous submission

- **Nothing identifying is emitted unless you set it**
- `blind` suppresses author, institution and logo even when set, so a deck flips to anonymous without editing its front matter

```
\usetheme[blind]{lucid}
```

snippets/blind.tex

Backup slides

- `backupframes` stops later slides inflating the page count

```
\backupframes

\begin{frame}{Backup: extra results}
  ...
\end{frame}
```

The next slide is a backup slide. Watch the page number restart.

References I

Richard D. McKelvey and Thomas R. Palfrey.
Quantal response equilibria for normal form games.
Games and Economic Behavior, 10(1):6–38, 1995.

Backup: the numbering restarted

- This is a backup slide
- The footer count started again at 1